

BAT Engineering

Hose Management System (HMS) 2.0

Functional Specification Document

User roles, functional modules, workflows and acceptance criteria

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1 Document control

Version	Date	Author	Summary of change
0.1	Jun 2026	Architecture	Initial System Design (draft, markdown)
0.2	Jun 2026	Architecture	Mobile & Offline Addendum A (draft, markdown)
1.0	27 June 2026	Architecture	Consolidated Functional Specification Document; review refinements incorporated

Supersession. This document consolidates and supersedes the earlier System Design and Mobile/Offline Addendum drafts. The field application described in those drafts as an offline *PWA* is superseded: the field client is delivered as a **packaged mobile app** built with Ionic + Angular + Capacitor (Angular running in a native shell), as defined herein. No competing approaches remain.

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2 Introduction

2.1 Purpose

This document defines what HMS 2.0 must do from a user and business perspective: the user roles, the functional modules (existing capabilities preserved and modernised, plus new capabilities), the key workflows, and testable acceptance criteria. It is read alongside the Technical Design Document, which defines how these requirements are implemented.

2.2 Scope and v1 boundaries

v1 delivers the web staff/admin console, the offline mobile inspector app, and the sync-ready backend, with a responsive (web) customer self-service portal. A dedicated customer mobile app, advanced analytics, external integrations and AI-assisted features are specified here but scheduled to later phases.

2.3 Intended audience

Product and engineering stakeholders, QA, and BAT business owners who need to confirm that HMS 2.0 preserves existing functionality and adds the agreed new capabilities.

2.4 Definitions

Terms (asset, WP/TP, grave/condemned date, certificate, packaged mobile app) are defined in the Technical Design Document, section 1.3.

3 System overview

HMS 2.0 manages the lifecycle of industrial hose assets for BAT and its customers. BAT staff use a web console to manage customers, products, reference data, inspections, certificates and scheduling. Inspectors use a packaged mobile app to perform inspections in the field, including offline. Customers use a responsive web portal to view their fleet, retest status and certificates. Because hoses are safety-critical and certificates serve as compliance evidence, the system emphasises data integrity, full history and verifiable certificates.

4 User roles and permissions

Role	Primary capabilities
System administrator	Full access including user and role management and all configuration
HMS administrator	Manage customers, products, reference data and retest scheduling
Inspector / assembly	Perform and edit inspections and capture results, online or offline
Reviewer	Review, approve or reject inspections and issue certificates
Customer user	View only their own customer's assets, retest status and certificates

5 Functional modules (existing, modernised)

Every legacy capability is preserved and improved.

5.1 Authentication and user management

Secure login (OIDC), password reset and optional multi-factor. Administrators invite and manage users and assign roles. Passwords are never displayed. Customer users are scoped to their own customer.

5.2 Customer management

Create, search, view and edit customers, including code, name, addresses, retest configuration and notes. Manage each customer's locations, named contacts and portal users. Customers are de-activated (soft-deleted), never destroyed.

5.3 Asset management

Maintain the hose inventory per customer with the full attribute set: serial, location, product, length, nominal bore, both end configurations (coupling, add-on, attach method, material for ends A and B), manufacture and condemned dates and notes. Filter and search assets, copy an asset, and view its inspection and certificate history. Assets carry a lifecycle state and a scannable tag.

5.4 Product catalogue

Manage products across categories (composite, PVC, rubber, stainless steel and others) with code, sub-category, type, grade, kind, test code, standard and an enabled flag, plus the working- and test-pressure values by nominal bore as a structured matrix.

5.5 Reference data administration

Manage the shared lookups - couplings, coupling add-ons, attach methods, materials, nominal bores and standards - as controlled lists referenced by assets and products (replacing free text).

5.6 Inspection management

Start a new-asset or service inspection for an asset, answer the applicable versioned question set, record pressure-test results and capture photos. Inspections are submitted for review; a reviewer approves or rejects. A certificate can be issued only from an approved inspection. Inspections may be created and completed offline on the mobile app and synced later.

5.7 Certificate management

Generate signed, versioned PDF/A certificates for approved inspections, individually or in bulk for a customer. Each certificate carries a verification token, can be downloaded, and retains version history. Bulk generation runs as a tracked background job.

5.8 Retest scheduling and reminders

Compute retest due dates per asset/customer and present due and overdue work. This replaces the manual Tests Due page and is augmented by the automated reminders below.

5.9 Search

Fast customer and asset search (by name, asset id or serial), with server-side filtering on the data grids.

6 New functional capabilities

6.1 Offline field inspection (packaged mobile app)

Inspectors download their assigned work, complete inspections, capture photos and pressure-test readings, and scan asset tags - fully offline - then sync when connectivity returns. First use must be online to authenticate and provision the device; subsequent use works offline within a bounded window.

6.2 Asset tagging and public certificate verification

Each hose carries a durable QR tag (RFID later). Scanning shows the asset's status and history. A public verification page lets anyone confirm a certificate is genuine, current and matches the asset.

6.3 Automated retest reminders and escalation

The system proactively emails and texts BAT and the customer ahead of retest due dates and escalates overdue assets, turning a passive list into a managed workflow.

6.4 Immutable audit and compliance ledger

Every change to a safety record is appended to a tamper-evident log, so BAT can demonstrate who inspected what and when during an audit or after an incident.

6.5 Verifiable digital certificates

Certificates are signed, versioned and verifiable, replacing static PDFs.

6.6 Customer self-service portal (responsive web in v1)

Customers view their fleet health, upcoming and overdue retests, and download certificates. Delivered as responsive web in v1; a customer mobile app is a later phase.

6.7 Fleet analytics and dashboards

Asset-health overview, failure-rate analysis by product/coupling/material, inspection throughput, and assets approaching end-of-life for proactive replacement.

6.8 Standards and specification validation

When an asset is built or inspected, the system validates the product, bore and coupling combination against pressure ratings and the applicable standard. Whether invalid configurations warn or block is an open decision.

6.9 Bulk import and export

Import customers and assets from spreadsheets and export data; this also serves the eventual legacy migration.

6.10 Integration API and webhooks (later phase)

Allow customer systems to retrieve asset and certificate data and to receive events such as certificate issued or asset condemned.

6.11 AI-assisted features (future phase)

Photo-based defect triage, natural-language fleet search and auto-drafted inspection notes - positioned as a later enhancement, not v1.

7 Key user workflows

7.1 Inspector - offline inspection lifecycle

- First launch (online): sign in, device is registered, in-scope data is downloaded to the device.
- In the field (offline): open the app, unlock with biometric/passcode, select an assigned asset, scan its tag, answer the inspection, record the pressure test and capture photos; data is saved locally.
- Submit the inspection locally; it is queued for sync.
- On reconnect (or manual Sync): queued inspections and photos are pushed; updates are pulled.
- A reviewer approves the inspection and a certificate is issued.

7.2 Administrator - onboard a customer and assets

- Create the customer with addresses, contacts, locations and retest configuration.
- Add assets, selecting product, bore, end configurations and tag; the system validates the configuration against the product's ratings.
- Optionally bulk-import assets from a spreadsheet.

7.3 Reviewer - review, approve and issue a certificate

- Open a submitted inspection and review answers, pressure-test results and photos.
- Approve (or reject with reason).
- Issue the certificate (single) or run a bulk generation for the customer.

7.4 Customer - verify a certificate

- Scan the hose's QR tag or open the verification link.
- The public page confirms the certificate is genuine, current and matched to the asset.

7.5 Retest reminder lifecycle

- The system computes due dates and sends advance reminders to BAT and the customer.
- Overdue assets are flagged and escalated.
- Due and overdue work appears on the dashboard for action.

8 Functional requirements and acceptance criteria

ID	Requirement	Acceptance criteria	Phase
FR-01	Secure authentication	User logs in via OIDC; invalid credentials are rejected; passwords are never displayed anywhere	1
FR-02	Role-based access	Each role sees and can do only what its permissions allow; a customer user cannot access another customer's data	1
FR-03	Customer CRUD	Create, search, view and edit customers; de-activation is a soft delete and the record remains in history	1
FR-04	Locations and contacts	Manage multiple locations and named contacts per customer	1
FR-05	Asset CRUD	Create and edit assets with full attributes and both end configurations from controlled lists; copy an asset	1
FR-06	Reference data	Manage couplings, add-ons, attach methods, materials, bores and standards; assets reference them by selection, not free text	1
FR-07	Product catalogue	Manage products and a WP/TP-by-bore matrix per product	1
FR-08	Search	Find customers and assets by name, id or serial with server-side filtering	1
FR-09	Asset lifecycle	Asset state reflects in-service, due, overdue, condemned or retired; a condemned asset never appears serviceable	1
FR-10	Inspection capture	Start new-asset or service inspections; answer the versioned question set; record pressure-test results and photos	2
FR-11	Review and approve	Inspections are submitted, then approved or rejected with reason; certificates issue only from approved inspections	2
FR-12	Certificate generation	Generate signed, versioned PDF/A certificates individually and in bulk; bulk runs as a tracked job	2
FR-13	Audit ledger	Every change to a safety record is recorded immutably and is tamper-evident	2
FR-14	Offline provisioning	First launch online registers the device and downloads in-scope data via the bootstrap snapshot	3

ID	Requirement	Acceptance criteria	Phase
FR-15	Offline operation	From the second launch the app works fully offline within the device's offline window after biometric/passcode unlock	3
FR-16	Sync triggers	Sync runs on app open, on reconnect and via a manual button; background sync is best-effort only	3
FR-17	Sync correctness	Local changes push with idempotency keys; pulled changes apply transactionally; no change is lost on partial sync	3
FR-18	Conflict handling	Submitted/approved inspections are server-authoritative; asset safety-field conflicts surface a screen and are never silently overwritten	3
FR-19	Device governance	An administrator can revoke a device; on next sync it is denied and instructed to wipe local data	3
FR-20	Asset tagging and scan	Each asset has a QR tag; scanning opens its status and history	3
FR-21	Public verification	Anyone can verify a certificate via its token without logging in	3
FR-22	Retest reminders	Advance and overdue reminders are sent by email and SMS to BAT and the customer	3
FR-23	Customer portal	Customers view their fleet, retest status and certificates (responsive web)	3
FR-24	Standards validation	Invalid product/bore/coupling combinations are detected; warn-or-block behaviour is configurable (decision pending)	4
FR-25	Bulk import/export	Customers and assets can be imported from and exported to spreadsheets with validation	4

9 Non-functional requirements (summary)

Area	Requirement
Availability	99.9% for the web console and API; the mobile app works fully offline
Performance	Interactive reads p95 under 300 ms; certificate generation is asynchronous
Offline	First launch online; subsequent launches offline within a bounded, server-governed window
Security	OIDC, RBAC, encryption in transit and at rest, encrypted on-device storage, secrets in a vault
Compliance	Full retention of safety/compliance history; tamper-evident audit; verifiable certificates
Data residency	Australian region (OCI Sydney)

Detailed non-functional design is in the Technical Design Document, sections 2 and 9-12.

10 Scope, assumptions and out-of-scope

10.1 Assumptions

- HMS is single-organisation (BAT) serving many customer accounts; customer users are strictly scoped to their own data.
- Development uses synthetic/dirty and sanitized sample data; full legacy migration is a later, controlled ETL.
- Inspectors carry managed mobile devices capable of biometric unlock.

10.2 Out of scope for v1

- Dedicated customer mobile app (responsive web portal only in v1).
- External system integrations and webhooks.
- AI-assisted features.
- Full production data migration (designed for, dry-run early, executed later).

11 Open questions

- Should standards validation warn or block on invalid configurations?
- What offline-window length balances field needs and security?
- How strict should asset-field conflict handling be for non-safety fields?
- Are open/reconnect/manual sync sufficient, or is true background sync required?
- Is the customer mobile app needed soon, or is the responsive portal sufficient for the foreseeable term?
- Expected photos per inspection and on-device retention policy?